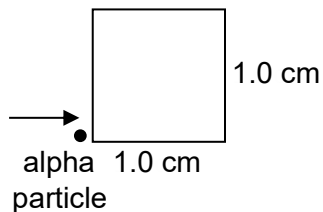


- 22 An alpha particle initially at rest is accelerated across a potential difference 250 V before entering a square uniform magnetic field of sides 1.0 cm with magnetic flux density 0.25 T at right angles as shown below.



What is the angle of deflection of the velocity of the alpha particle?

A 0°

B 33°

C 51°

D 180°

By law of conservation of energy,
loss in EPE = gain in KE

$$(2e)V = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{4eV}{m}}$$

$$= 2\sqrt{\frac{eV}{m}}$$

Magnetic force provides centripetal force, by Newton's second law of motion,

$$F_{\text{net}} = ma$$

$$F_c = ma_c$$

$$B(2e)v = \frac{mv^2}{r} \quad (\because \sin \theta = 1)$$

$$r = \frac{mv}{2Be}$$

$$= \frac{1}{B} \sqrt{\frac{mV}{e}}$$

$$= \frac{1}{0.25} \sqrt{\frac{4(1.66 \times 10^{-27})(250)}{1.60 \times 10^{-19}}}$$

$$= 1.3 \text{ cm}$$

Let the angle of deflection be α ,

$$r \sin \alpha = d$$

$$\alpha = \sin^{-1} \frac{d}{r}$$

$$= \sin^{-1} \frac{1.0}{1.3}$$

$$= 51^\circ$$

Ans: C